PRD — AiversePulse Enterprise: growth since the preliminary round

Between the Phase 1 proposal (24 Oct 2026) and the final submission (1 Nov 2026), AiversePulse grew from a reskilling prototype into an organization-ready platform. We kept the Phase 1 core and built on it in three directions: plans that actually teach, tools for HR to manage a whole workforce, and the accounts, security, and infrastructure an organization needs to adopt it.

Where we started

The Phase 1 proposal already described the foundation, which we do not present as new:

- A LangGraph reskilling pipeline (supervisor, profiler, knowledge search, market analyst, gap analyst, pathway, coach, reviewer).
- Role matching across about 3,000 ESCO v1.2 occupations, with a requirement model trained offline on O*NET 31.0.
- An STF-KAN rising-skill outlook, a daily AI-trend board, and SDG links to the 169 UN targets.
- Two AI scenarios (assistive and agentic) and a 30/60/90-day plan capped to weekly hours.
- Resume screening with plain-language thresholds and blind review, a RAG chat agent, and a Three.js 3D mock interviewer.
- A Replit frontend, a dual-node VPS backend, SQLite and Chroma storage.

At a glance

Area	Phase 1 (24 Oct)	Final (1 Nov)
Learning plan	Activities with cited sources, capped to weekly hours	Full lessons: instructions, exercises, deliverables, knowledge checks, completion criteria; progress tracking
HR team view	Run analyses per employee, approve plans	Workforce skills-gap heatmap by department, role, dataset or group; team-plan drafts
Plan governance	Draft until named approval; immutable versions	Review queue with search, filters, archive, soft delete, bulk actions; confirmed status changes
Hiring	Openings, thresholds, blind review	Posting agent to six social platforms; HR operator chat; voice job drafts
Organisations	Employee and HR accounts	Companies and schools, invite-by-email members, profiles, monthly subscription
AI models	Optional Gemini or gateway	Admin allow-list; model per task, per organisation and per member; usage billed
Assistant	RAG chat	Long-term memory per account, saved chats, live web search tools

Area	Phase 1 (24 Oct)	Final (1 Nov)
Security	Session cookies, CSRF header, document guard	Email-link verification, slider check, risk scoring, login Bloom filter, self-service deletion
Storage	SQLite + Chroma	PostgreSQL primary/standby with pgvector across nodes; SQLite kept as local fallback

1. Learning plans that teach

In Phase 1 a plan listed cited activities; now each activity is a small lesson the learner can work through. Every item carries practical instructions, a worked example, an exercise, an observable deliverable, a knowledge check and completion criteria. Plan content streams into the report as it is generated, and learners can mark progress item by item.

The hard part was mixing AI-written lessons with predictable planning rules. We solved it with structured schemas, citation-ID checks, per-phase hour limits and bounded repairs:

- **Time fits the learner.** Phase capacity = weekly hours × 30 ÷ 7. At 6 hours a week that is 25.7 hours per phase, or 77.1 hours over 90 days. Skills that do not fit are listed as deferred.
- **Sources stay real.** External resources come only from retrieved evidence. The Reviewer rejects any citation ID that was not in the retrieved set.
- **Failures are visible.** The Reviewer allows at most two targeted repairs; anything still failing is flagged for a person, not hidden.
- **Fallback lessons.** When the learning-designer model is unavailable, a built-in catalogue of concept lessons (for example visual hierarchy, colour and contrast, spacing and alignment) keeps the plan useful instead of inserting a skill name into generic advice.

Changing weekly hours, skills or the AI scenario now reuses the unaffected parts of the analysis, so a learner can adjust their plan without rerunning the whole pipeline.

2. Workforce skills-gap heatmap

HR advisors can now see a whole organisation's gaps on one screen instead of opening plans one by one. Under HR / Enterprise → Overview, the heatmap compares role-specific assessments across departments, roles, uploaded datasets, external groups or sources.

- Filters cover department, role, skill, skill category and AI scenario; HR can inspect any assessed cell.
- Selected gaps can be saved as a separate team-plan draft for HR review. Existing individual plans are never changed.
- Privacy guards: a group must have at least 3 people and 60% assessment coverage before it is shown, and views show aggregates only, never rankings of people.
- A clearly labelled preview uses 40 fictional employees across five departments and seven skills (such as AI literacy, data analysis and workflow automation). It is never mixed into real analytics.
- The heatmap loads without a new model call.

New HR analyses also reserve persistent employee IDs (E-001, E-002 and so on) that are never reused, even if a run fails or its plans are deleted.

3. HR review queue and plan governance

Phase 1 had named approval and immutable versions; the final build adds the day-to-day tools to manage many plans.

- **Find and filter.** Search by name, contact details, employee ID or target role; filter by Pending, Approved, Rejected or Archived.
- **Manage.** Edit candidate details and reviewer notes, archive or restore, and soft-delete after confirmation. Bulk actions handle up to 50 reviews per request, and each edit carries a revision number so stale saves are rejected.
- **Plan status.** An editable Draft / Flagged / Approved column asks for confirmation, then writes a new immutable plan version and an audit event while keeping the AI report. Approval still requires a Draft plan.
- **Audit kept.** Soft-deleted reviews leave reports, plan versions and the append-only HR audit trail in the database. HR review status is separate from plan approval, so editing a review never approves or changes the AI-generated plan.
- **No manual migration.** The new tables and triggers are created automatically on start-up.

4. Hiring reach: posting agent and HR operator

Phase 1 screened applicants who arrived through a public link; the final build also brings applicants in.

- **Hiring plugins.** HR connects company accounts once: a LinkedIn company page, Facebook Page and X account through OAuth, a Telegram channel through a bot, and Discord and Slack through webhooks. A posting agent writes a post suited to each platform and shares every new opening to all of them.
- **Tokens stay safe.** Platform tokens are encrypted at rest and never returned to the browser. A platform without app keys is shown as "needs app keys" instead of failing silently.
- **HR operator.** HR can type or say a request such as "send the nurse opening to our Telegram". The chat replies with a proposal naming the opening and accounts, and nothing is posted until the advisor clicks it, through the same endpoint as the Post button.
- **Voice.** Openings can be drafted from a spoken description, and HR can ask questions about their team by voice or text.

5. Enterprise accounts, billing and model control

To let a real organisation adopt the product, we added the account structure around it.

- **Companies and schools.** An HR advisor's organisation is either a company (members are employees) or a school (members are students), so career centres can use the same tools.
- **Invite only.** HR invites members by email; there is no public sign-up page for organisation members. The invite link (valid 7 days) opens a "set your password" page, and HR can disable or remove members later.
- **Full profiles.** Every account has a name, phone, title, department and photo.
- **Plans and payment.** Individuals choose Free (Quick Match, AI trends, 3 new plans a month, read aloud) or Pro (unlimited plans and chat, resume analysis, mock interviews). Organisations pay a flat monthly Enterprise subscription through Stripe that renews automatically.
- **Model control.** The platform admin keeps an allow-list of AI models. Each organisation picks a model per task (coaching, resume screening, job posts), and HR can override it for an individual

member. The admin sees each organisation's monthly bill: subscription payments plus everyone's AI usage.
- **More than one site.** The admin registers allowed frontend sites. Every site serves the same app against one backend, and sign-in works across domains with a Bearer token where browsers block third-party cookies.

6. A personal assistant that remembers

The Phase 1 RAG chat answered one question at a time. The assistant now keeps context across sessions.

- **Long-term memory.** The agent saves short facts about the person (up to 60, with at most 3 new facts per reply) and carries them into later chats. Each account only sees its own facts, and the person can list, delete or clear them in Settings.
- **Saved conversations.** Chat history is kept per account so learners can return to earlier threads.
- **Live tools.** While answering, the agent can search the web and read pages, alongside the app's own MCP tools for ESCO occupations, skill outlook, the trend board and role matching. Web text is treated as untrusted data, never as instructions.

7. Account security and privacy

Opening sign-up to the public meant protecting accounts as well as resumes.

- **Email-link verification.** Sign-up and password reset send one link. Opened in the same browser, it continues there; opened elsewhere, it shows a 6-digit code to type into the original browser. Links are single-use, expire after 10 minutes, allow 5 wrong tries, and only hashes are stored.
- **Slider check.** Sign-up needs a server-drawn slider puzzle that also checks the drag looks human. The gap position exists only in the image, never as a number sent to the browser.
- **Risk scoring.** Every sign-up, login and payment is scored 0–100 from device, behaviour and velocity signals. Scores of 40 or more are flagged for an admin; 70 or more are blocked. IP addresses are stored only as salted hashes.
- **Login Bloom filter.** Unknown usernames get exactly the same response and timing as a wrong password, so attackers cannot tell which accounts exist.
- **Self-service deletion.** One transaction deletes the person's login, plans, chats and interview records, and anonymises the records we must keep, such as payments and security events. The Privacy Policy describes exactly this behaviour.

8. Infrastructure for more than one node

The Phase 1 backend stored everything in SQLite on the main node. The final build adds PostgreSQL with pgvector for shared records and knowledge retrieval across application nodes, with primary/standby compose files, a watchdog, scheduled backups and TLS. Schemas are applied through a migration flow (the 29 Oct enterprise migration added organisations, subscriptions, per-member models, agent memory and chat history). SQLite stays as the local fallback, so a fresh clone still runs without a managed database.

Impact and KPI status

The secondary KPI held; the primary KPI is enforced in code but has not yet been measured in aggregate.

KPI (from Phase 1)	Phase 2 target	Status at final submission
Plan activities passing the evidence check on first draft	≥ 95%	Enforced by the Reviewer (citation IDs, gap links, positive hours, phase limits); aggregate rate not yet measured
Activities shown for approval that are verified or flagged	100%	Built into the Reviewer (unresolved errors flag the plan for a person); not yet measured in aggregate
Rising-skill lift over base rate	≥ 1.39×	1.389× over a 2.09% base rate on held-out data (historical result)

The upgrades support SDG 8.5 (decent work) and SDG 4.4 (skills for employment) by giving learners reviewable plans that fit their time, and giving organisations evidence for training decisions. We have not yet shown changes in employment, wages or completed learning; that needs a pilot.

What comes next

- ☐ Measure first-draft evidence compliance on a fixed set of generated plans.
- ☐ Run a supervised pilot with a university, TVET provider or employer to measure plan usefulness, completion and cost per plan.
- ☐ Add OCR for image-only resumes and detect text hidden behind shapes.
- ☐ Validate role matching against real career-transition outcomes.
- ☐ Support ESCO's 28 languages for non-English users.
- ☐ Add per-agent model routing with cost tracking, and a shared rate-limit store and job queue for multi-worker deployments.